

Let us now examine three potential trading strategies represented by x , y , and z . The expected market impact and corresponding timing risk for these strategies and their consequence on the overall risk-return profile for optimal portfolio A1 is shown in Table 10.1.

Notice the reduction in return is equal to the trading cost corresponding to the strategy but the increase in risk is actually less than the timing risk of the strategy. This is because risk is sub-additive (the variance expression is additive making the square root of this term less than additive). For example, to implement portfolio A1 using strategy y returns are expected to decline by 0.10% (10 bp) and risk will increase by 0.224% (22.4 bp) due to the corresponding timing risk of the transaction. Notice that the overall risk consequence from the strategy is much less than the risk incurred on the day of 3% (300 bp) due to the sub-additive nature of risk. Variance is additive and risk is sub-additive (Equation 10.6). On the other extreme, let us evaluate implementation via strategy z . Here return will be reduced by 2% due to the market impact of strategy z and the increase in portfolio risk will be negligible at 0.002% or 0.2 bp. Notice that for the investment portfolio the underlying market impact cost of the strategy has a much more dramatic effect on the ex-ante portfolio than the timing risk of the strategy. This is an important observation for the portfolio manager when devising the appropriate strategy to execute the trade.

Therefore, even a strategy with a large quantity of timing risk will have a much smaller effect on overall portfolio risk. But a strategy with a large quantity of market impact cost will have a large effect on overall portfolio returns.

Following the example in Table 10.1, the best the manager can expect to do after trading costs is to realize an ex-post portfolio return from 8 to 9.90%, with corresponding portfolio risk of 20.002 to 20.224%,

Table 10.1 Cost-Adjusted Risk-Return Values

Scenario	Trading Costs		Adjusted Risk-Return	
	Impact	Timing Risk	Return	Risk
Portfolio A1			10%	20%
Strategy y	0.10%	3.00%	9.90%	20.224%
Strategy x	0.25%	0.50%	9.75%	20.006%
Strategy z	2.00%	0.25%	8.00%	20.002%